

Disciplina: Eletrônica Básica

Aula_5 - Diodos

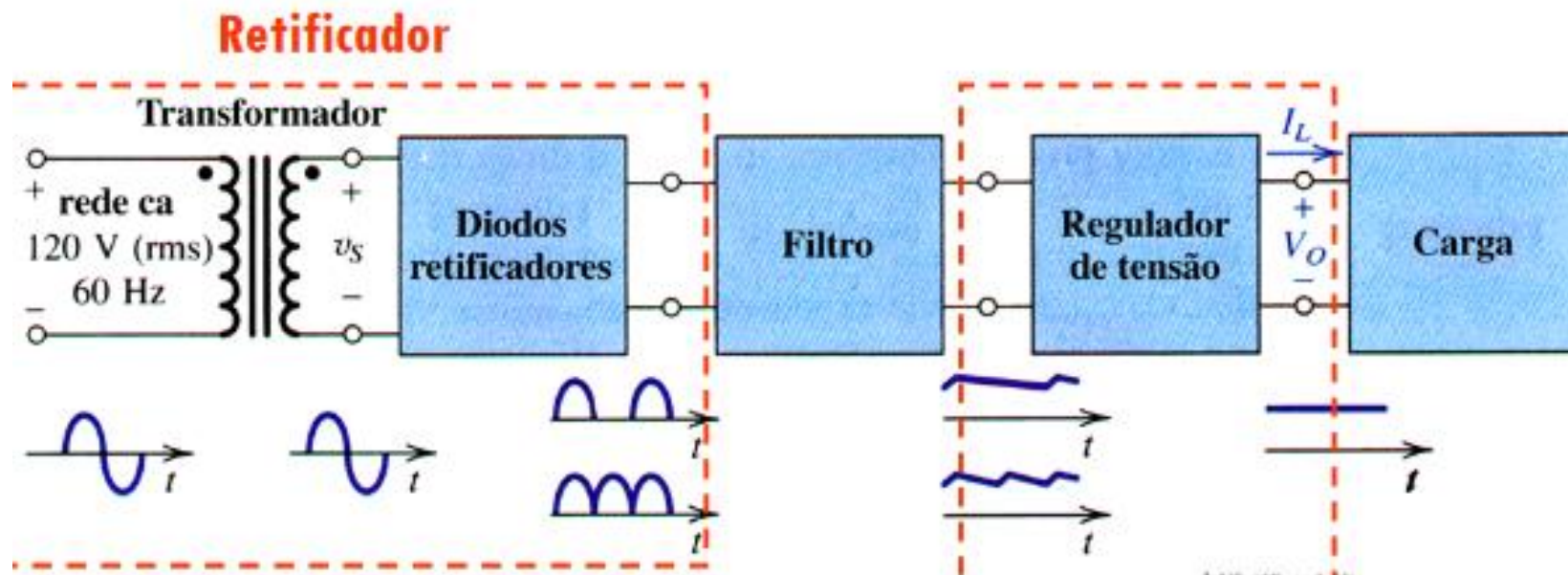
Ambiente Blackboard

Apresentação

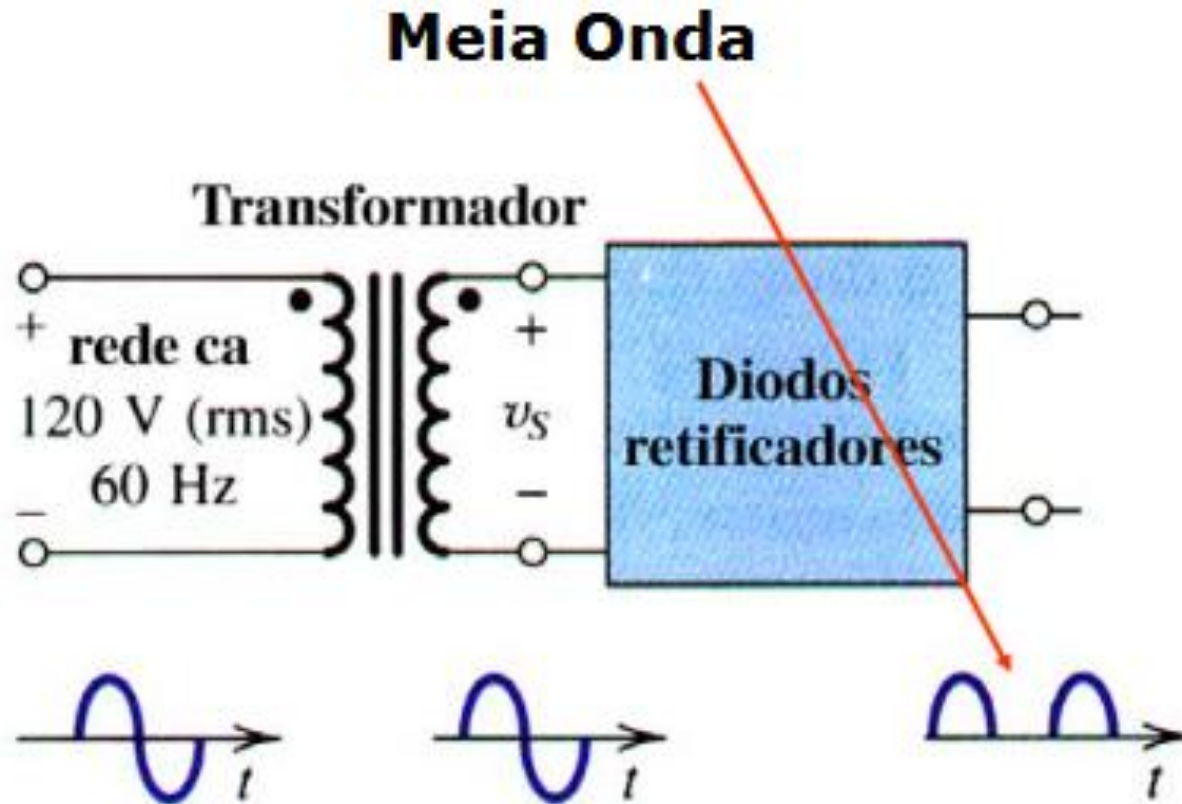
- 1) Meia onda - Exercícios
- 2) Onda completa CT - Exercícios
- 3) Onda completa Ponte – Exercícios
- 4) Filtro capacitivo
- 5) Exercícios - Tarefas

Retificadores

- Circuitos utilizados para conversão de CA/CC

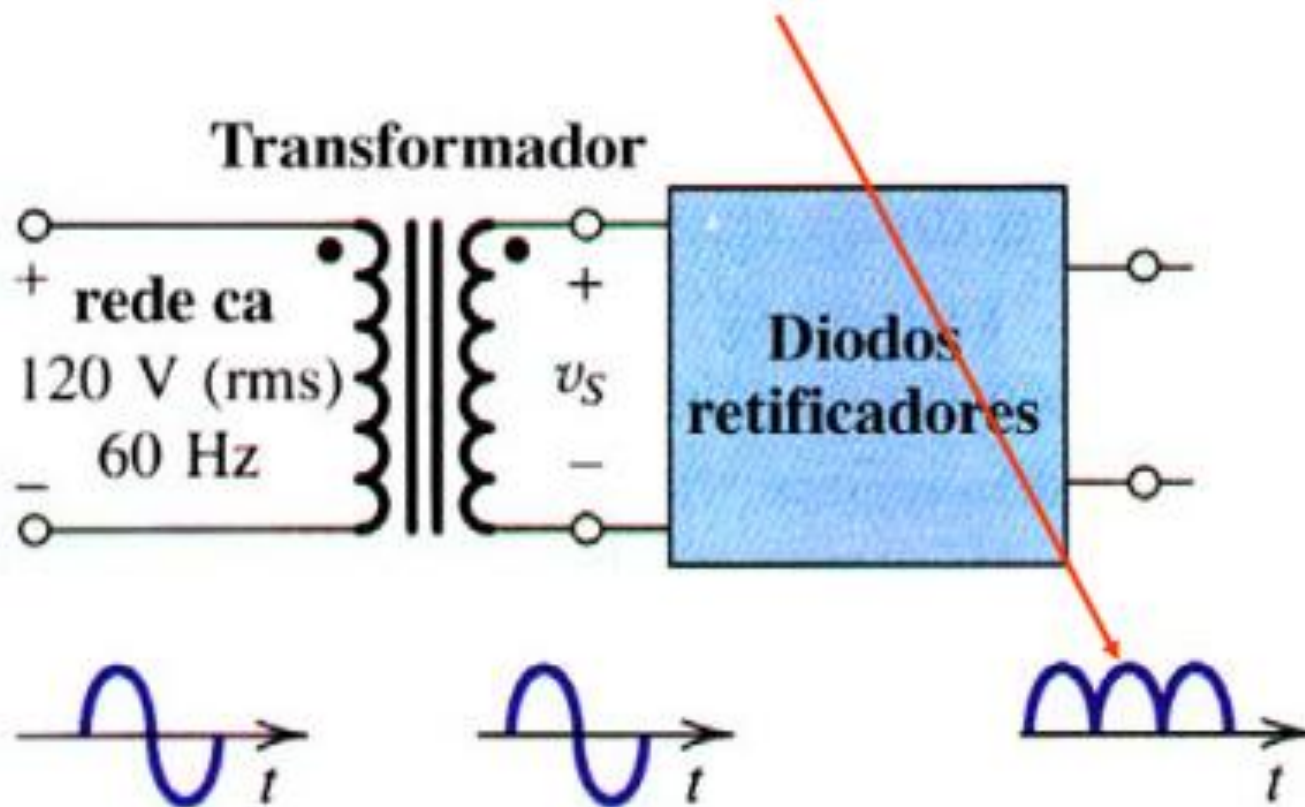


Retificador – Meia Onda



Retificador – Onda Completa

Onda Completa



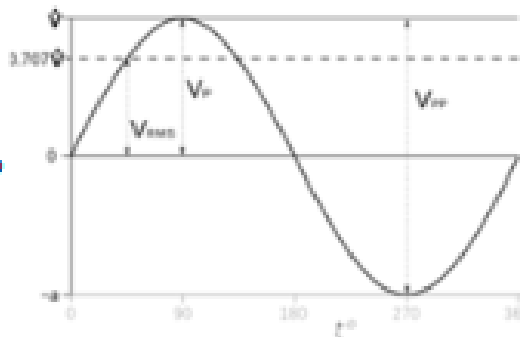
Revisão – Ondas Senoidais

Ondas Senoidais

$V_p = \hat{V}$ = valor de pico

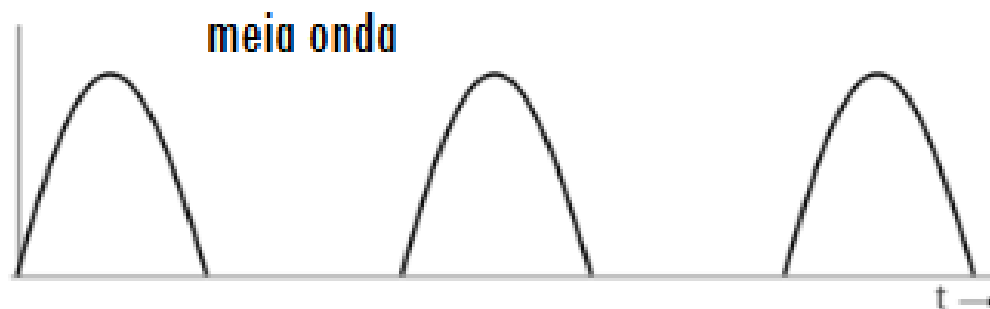
V_{DC} ou V_m = valor DC ou valor médio

V_{rms} = valor quadrático médio (rms)



$$V_{DC} = \frac{1}{T} \int_0^T V_p \sin(\omega t) dt = 0$$

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T [V_p \sin(\omega t)]^2 dt} = \frac{V_p}{\sqrt{2}}$$



$$V_{DC} = \frac{1}{T} \left[\int_0^{T/2} V_p \sin(\omega t) dt + \int_{T/2}^T 0 dt \right] = \frac{V_p}{\pi}$$

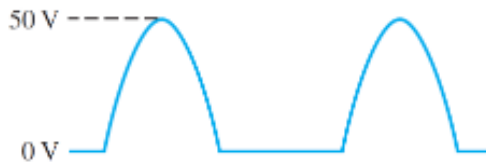
$$V_{rms} = \sqrt{\frac{1}{T} \int_0^{T/2} [V_p \sin(\omega t)]^2 dt} = \frac{V_p}{2}$$

Retificador – Meia Onda

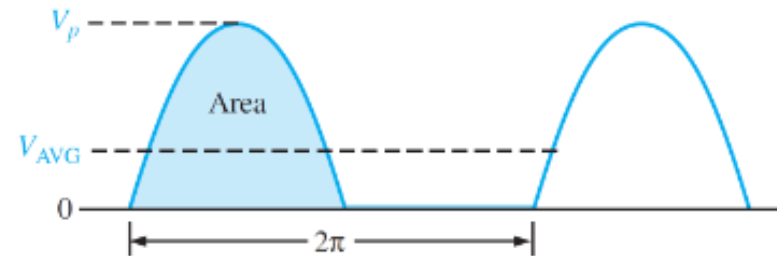
Average Value of the Half-Wave Output Voltage

- The average value of the half-wave rectified output voltage is **the value you would measure on a dc voltmeter**.
- Mathematically, it is determined by finding **the area under the curve over a full cycle**, and then dividing by 2π (the number of radians in a full cycle).

$$V_{AVG} \text{ or } V_{dc} = \frac{V_P}{\pi} = 0.318 V_P$$



$$V_{AVG} = \frac{V_P}{\pi} = \frac{50 \text{ V}}{\pi} = 15.9 \text{ V}$$



Notice that V_{AVG} is 31.8% of V_P .

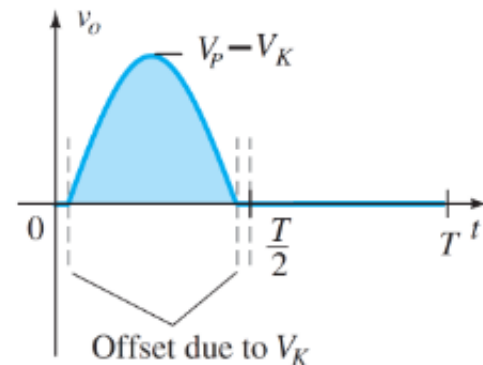
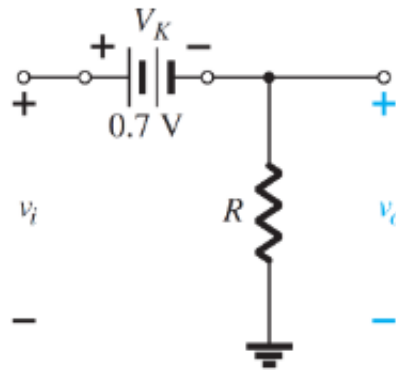
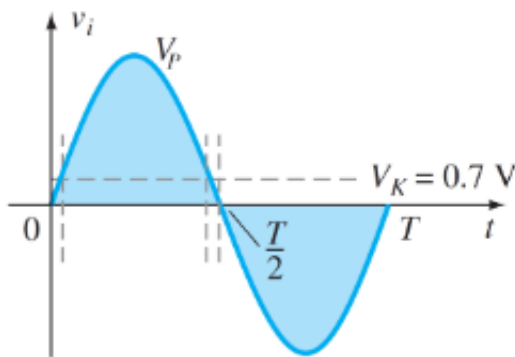
Retificador – Meia Onda

Effect of the Barrier Potential on the Half-Wave Rectifier Output (Using the practical diode model)

- When $V_{in} < 0.7$ the diode is OFF (O.C.) and ($V_{out} = 0V$)
- When $V_{in} > 0.7$ the diode is ON and ($V_{out} = V_{in} - 0.7$)
- The peak output ($V_{P(out)} = V_{P(in)} - 0.7$)

When $V_P \gg 0.7 \rightarrow V_{AVG} \approx \frac{V_P - 0.7}{\pi}$

$V_K = 0.7 V$



Retificador – Meia Onda

Half-wave rectifier with transformer coupled input voltage.

$$n(\text{Turns ratio}) = \frac{N_{sec}}{N_{pri}}$$

$$V_{sec} = nV_{pri}$$

If $n < 1$ **step down** transformer and $V_{sec} < V_{pri}$

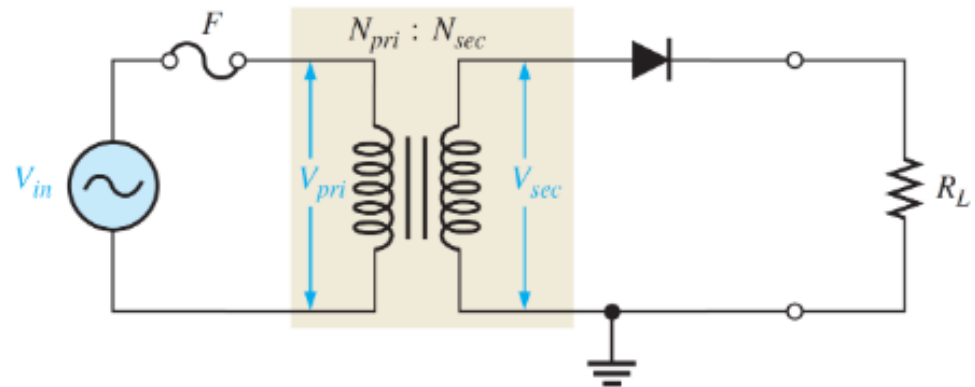
If $n > 1$ **step up** transformer and $V_{sec} > V_{pri}$

The peak output of the rectifier is:

$$V_{p(out)} = V_{p(sec)} - 0.7$$

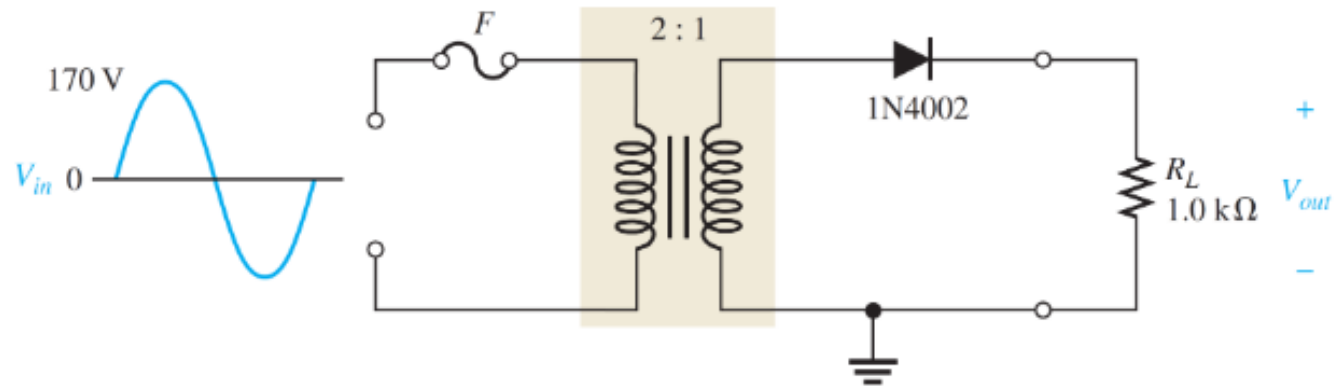
and

$$PIV = V_{p(sec)}$$



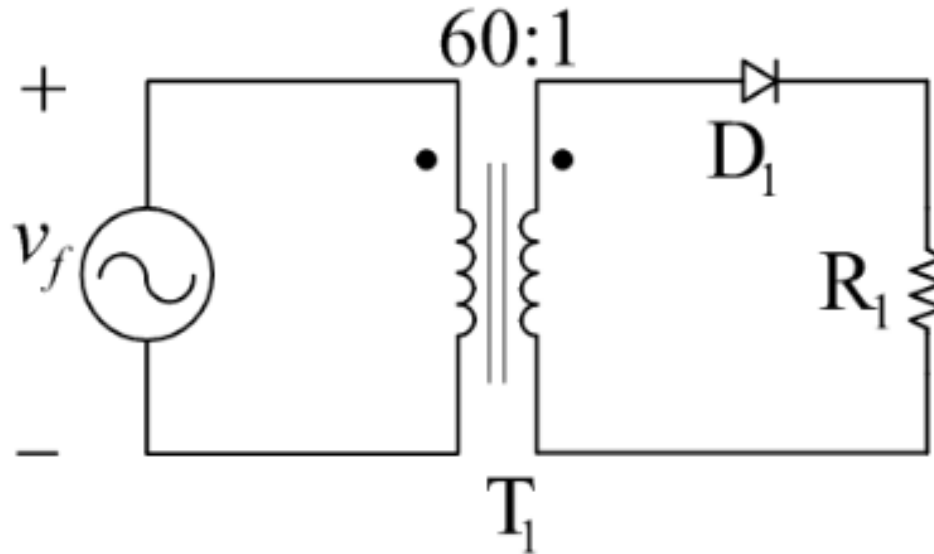
Retificador – Meia Onda

EXAMPLE 3 Determine the peak value of the output voltage.



- $n = \frac{1}{2} = 0.5$
- $V_{p(sec)} = nV_{p(pri)} = 0.5 \times 170 = 85 \text{ V}$
- $V_{p(out)} = V_{p(sec)} - 0.7 = 84.3 \text{ V}$
- $PIV = V_{p(sec)} = 85 \text{ V}$

Meia onda - Exercícios



Considerando os dados ao lado, determine:

- Tensão eficaz no primário de T_1 ;
- Tensão eficaz no secundário de T_1 ;
- Tensão média na saída;
- Tensão de pico na saída;
- Tensão reversa sobre o diodo;
- Corrente média na saída.

$$v_f(t) = 311 \cdot \text{sen}(377 \cdot t) \text{ V};$$

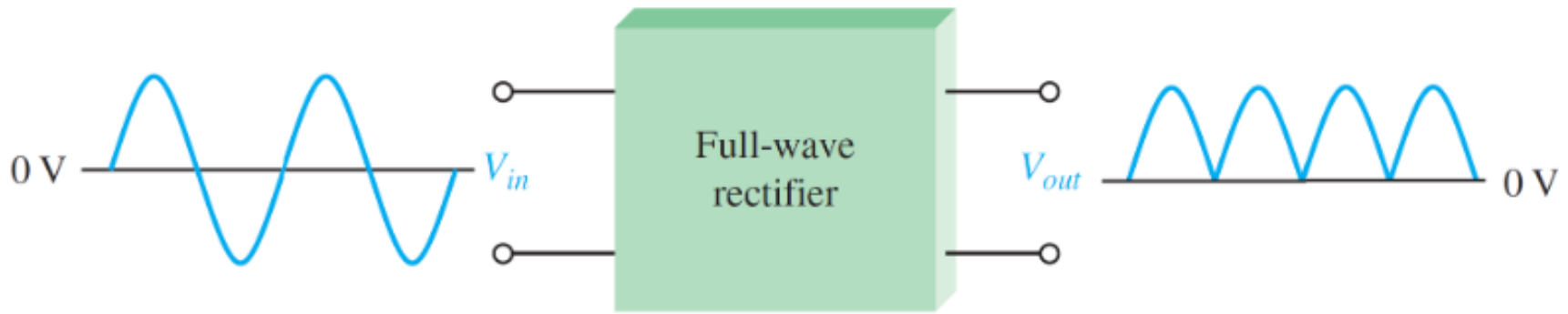
$$R_1 = 5 \, \Omega;$$

$$D_1 = \text{ideal};$$

$$T_1 = 60 : 1.$$

Retificador - Onda Completa

- A **full-wave rectifier** allows unidirectional (one-way) current through the load during the entire cycle of the input, whereas a half-wave rectifier allows current through the load only during one-half of the cycle.
- **FWR** : $f_{out} = 2 f_{input}$ [twice the input frequency]
- $V_{AVG} = \frac{2V_P}{\pi} = 0.637 V_P$ [twice that of HWR]



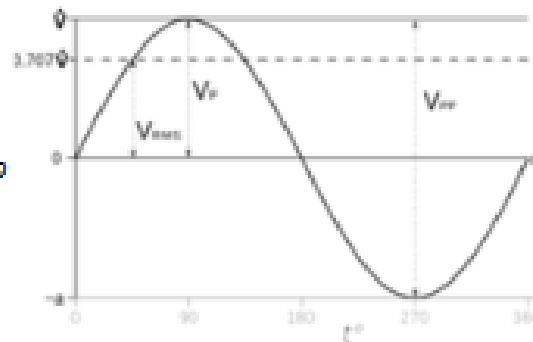
Retificador - Onda Completa

Ondas Senoidais

$V_p = \hat{V}$ = valor de pico

V_{DC} ou V_m = valor DC ou valor médio

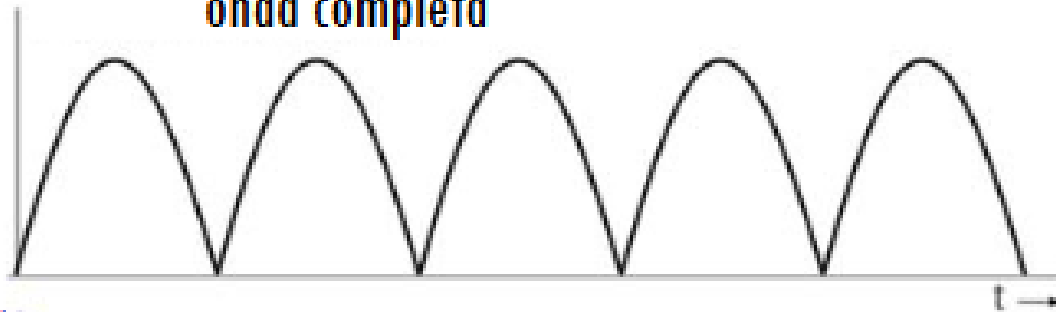
V_{rms} = valor quadrático médio (rms)



$$V_{DC} = \frac{1}{T} \int_0^T V_p \sin(\omega t) dt = 0$$

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T [V_p \sin(\omega t)]^2 dt} = \frac{V_p}{\sqrt{2}}$$

onda completa

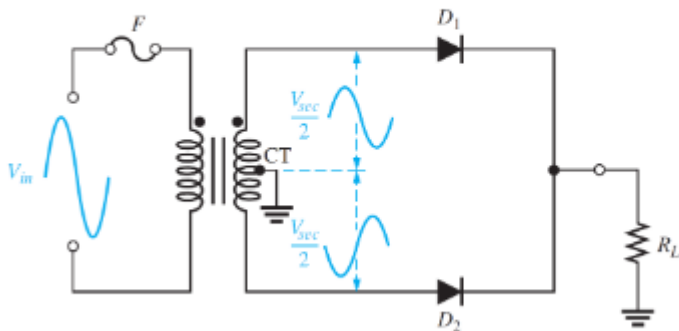


$$V_{DC} = 2 \times \frac{V_p}{\pi}$$

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T [V_p \sin(\omega t)]^2 dt} = \frac{V_p}{\sqrt{2}}$$

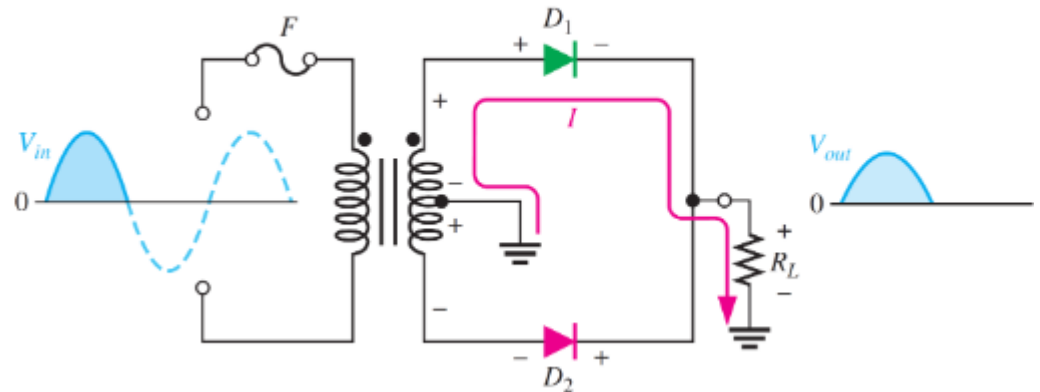
Onda completa Center Tap (CT)

Center-Tapped Full-Wave Rectifier (CT-FWR)

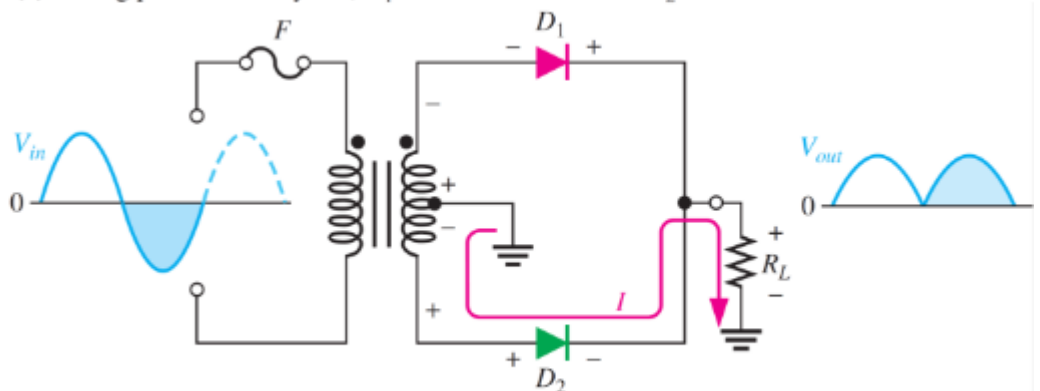


- During the **+ve** half cycle **D1 is ON** and **D2 is OFF**
- During the **-ve** half cycle **D1 is OFF** and **D2 is ON**
- The current passes through R in the same direction in both half cycles

$$V_{out} = \frac{V_{sec}}{2} - 0.7$$



(a) During positive half-cycles, D_1 is forward-biased and D_2 is reverse-biased.



(b) During negative half-cycles, D_2 is forward-biased and D_1 is reverse-biased.

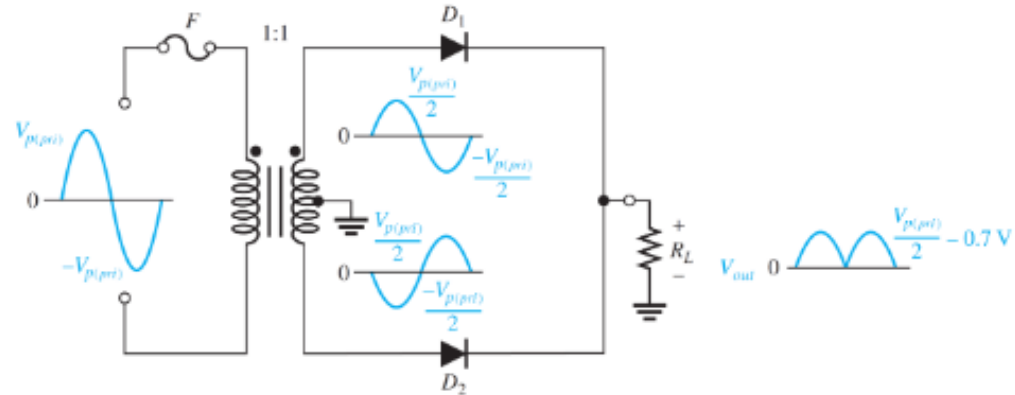
Onda completa Center Tap (CT)

Effect of the Turns Ratio on the Output Voltage

- $n = 1$
- The total secondary voltage is:

$$V_{p(sec)} = V_{p(pri)}$$

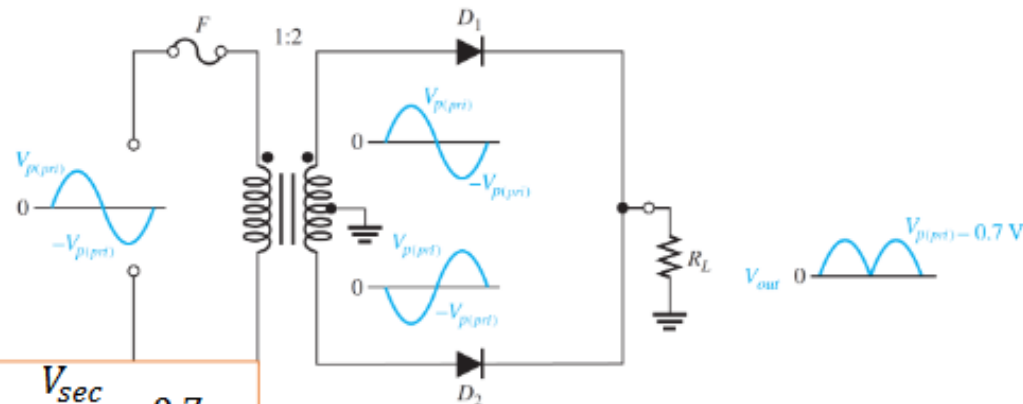
$$\begin{aligned} \bullet V_{p(out)} &= \frac{V_{p(sec)}}{2} - 0.7 \\ &= \frac{V_{p(pri)}}{2} - 0.7 \\ &= \frac{V_{p(input)}}{2} - 0.7 \end{aligned}$$



- $n = 2$
- The total secondary voltage is:

$$V_{p(sec)} = 2 V_{p(pri)}$$

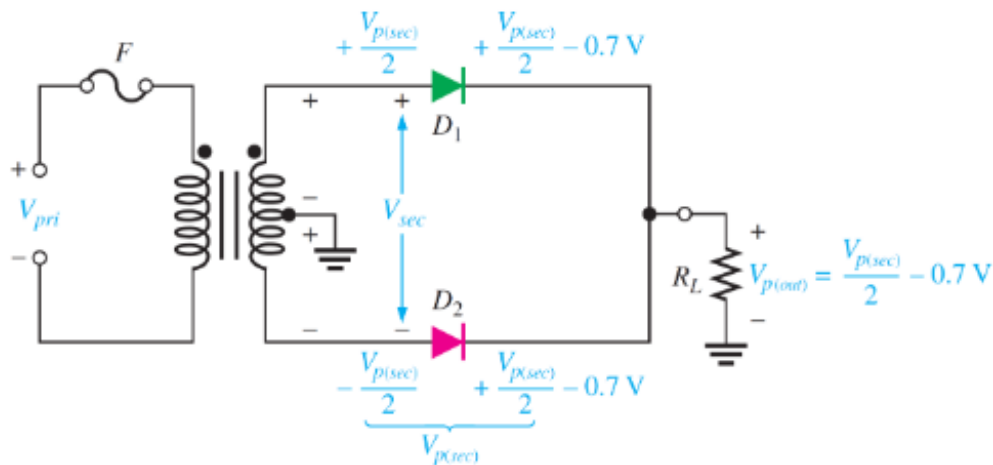
$$\begin{aligned} \bullet V_{p(out)} &= \frac{V_{p(sec)}}{2} - 0.7 \\ &= V_{p(pri)} - 0.7 \\ &= V_{p(input)} - 0.7 \end{aligned}$$



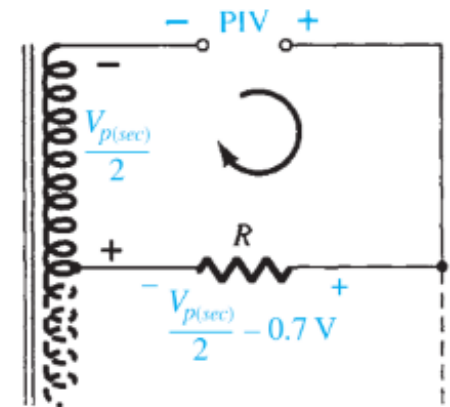
$$V_{out} = \frac{V_{sec}}{2} - 0.7$$

Onda completa Center Tap (CT)

Peak Inverse Voltage



$$\begin{aligned} PIV &= \left(\frac{V_{p(sec)}}{2} - 0.7 \text{ V} \right) - \left(-\frac{V_{p(sec)}}{2} \right) = \frac{V_{p(sec)}}{2} + \frac{V_{p(sec)}}{2} - 0.7 \text{ V} \\ &= V_{p(sec)} - 0.7 \text{ V} \end{aligned}$$



$$\begin{aligned} PIV &= \frac{V_{p(sec)}}{2} + \frac{V_{p(sec)}}{2} - 0.7 \text{ V} \\ &= V_{p(sec)} - 0.7 \text{ V} \end{aligned}$$

$$\boxed{PIV = V_{p(sec)} - 0.7}$$

Onda completa Center Tap (CT)

Example 4

- (a) Show the voltage waveforms across each half of the secondary winding and across R_L
- (b) Find the DC value of the output.
- (c) What minimum PIV rating must the diodes have?

Solution (a) $n = \frac{1}{2} = 0.5$

- The total peak secondary voltage is:

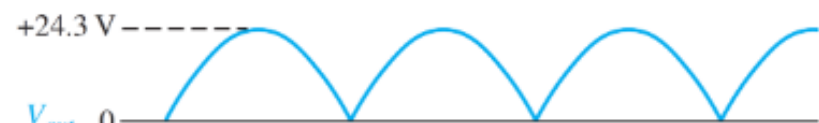
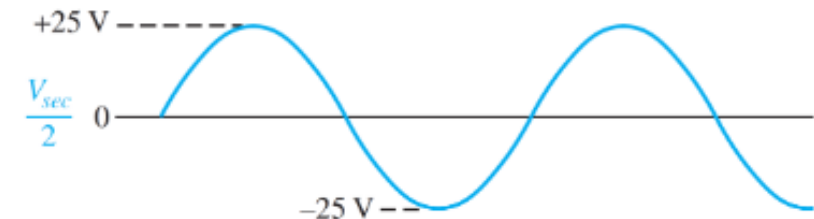
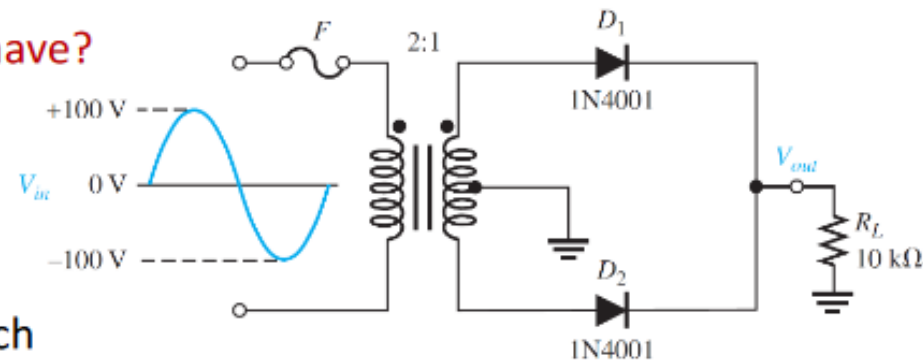
$$V_{p(sec)} = nV_{p(pri)} = 0.5 \times 100 = 50 \text{ V}$$
- Hence There is a $\frac{V_{p(sec)}}{2} = 25 \text{ V}$ peak across each half of the secondary with respect to ground.
- The load voltage has a peak value:

$$V_{p(out)} = \frac{V_{p(sec)}}{2} - 0.7 = 24.3 \text{ V}$$

(b) $V_{AVG} = \frac{2V_{P(out)}}{\pi} = 0.637 V_{P(out)} = 15.48 \text{ V}$

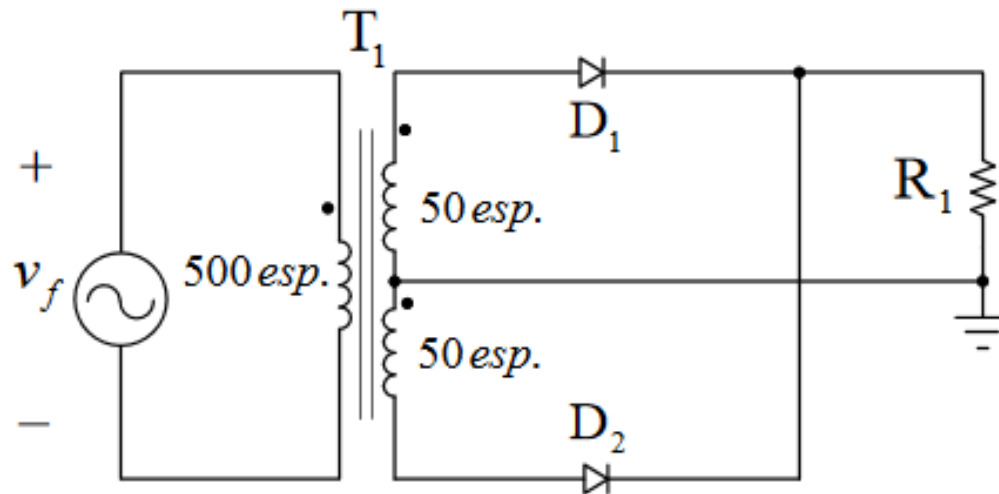
- (c) Each diode must have a minimum PIV rating of

$$PIV = V_{p(sec)} - 0.7 = 49.3 \text{ V}$$



Onda completa CT - Exercício

Considerando o circuito abaixo:



Considerando os dados ao lado, determine:

- Tensão eficaz no primário de T_1 ;
- Tensão eficaz no secundário de T_1 ;
- Tensão média na saída;
- Tensão de pico na saída;
- Tensão reversa sobre os diodos;
- Corrente média na saída.

$$v_f(t) = 311 \cdot \sin(377 \cdot t) V;$$

$$R_1 = 5 \Omega;$$

$$D_{1_2} = \textit{ideais};$$

$$T_1 = \begin{cases} 10:1 \\ 10:1 \end{cases}$$

Tarefas

- Simule os circuitos apresentados nessa aula.

Onda completa Ponte

Bridge Full-Wave Rectifier

During the positive half-cycle

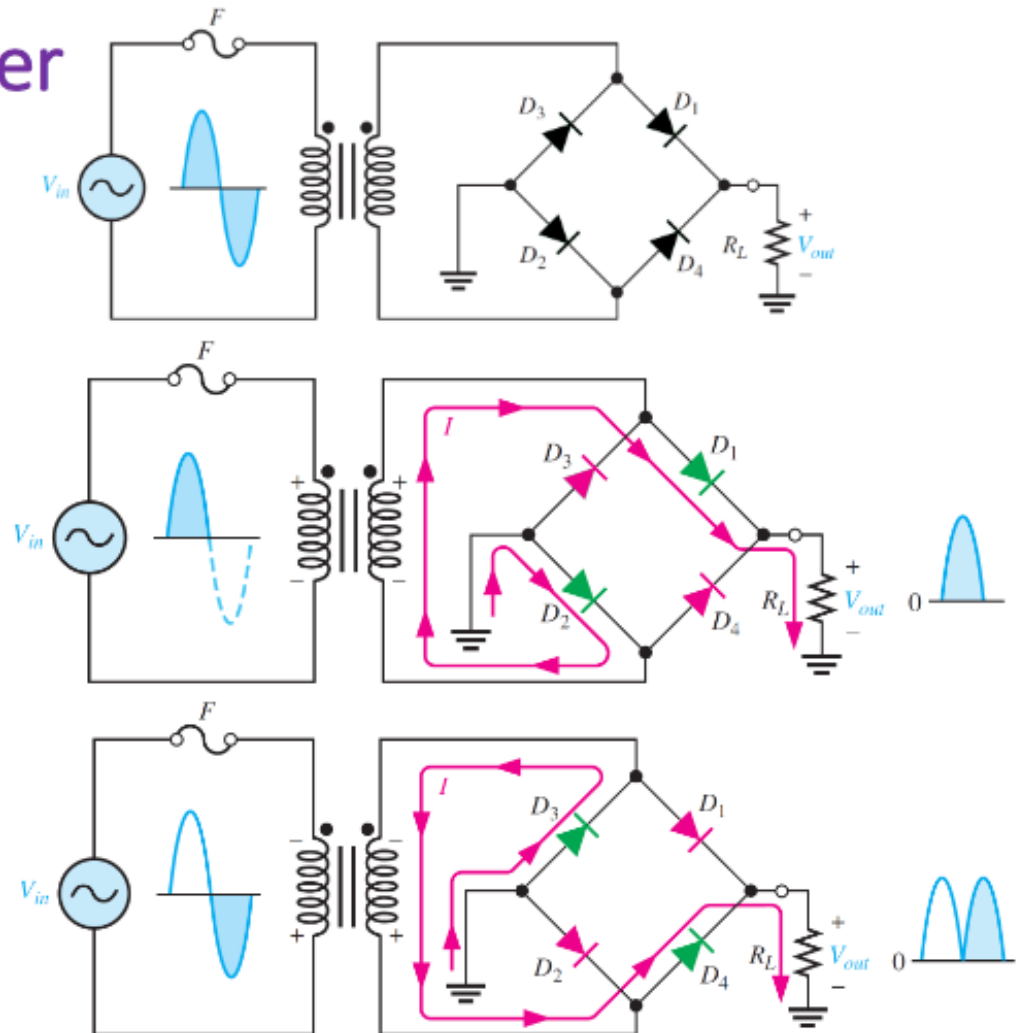
- D_1 and D_2 are forward-biased and conduct current.
- D_3 and D_4 are reverse-biased.

During the negative half-cycle

- D_3 and D_4 are forward-biased and conduct current.
- D_1 and D_2 are reverse-biased.

Practical:

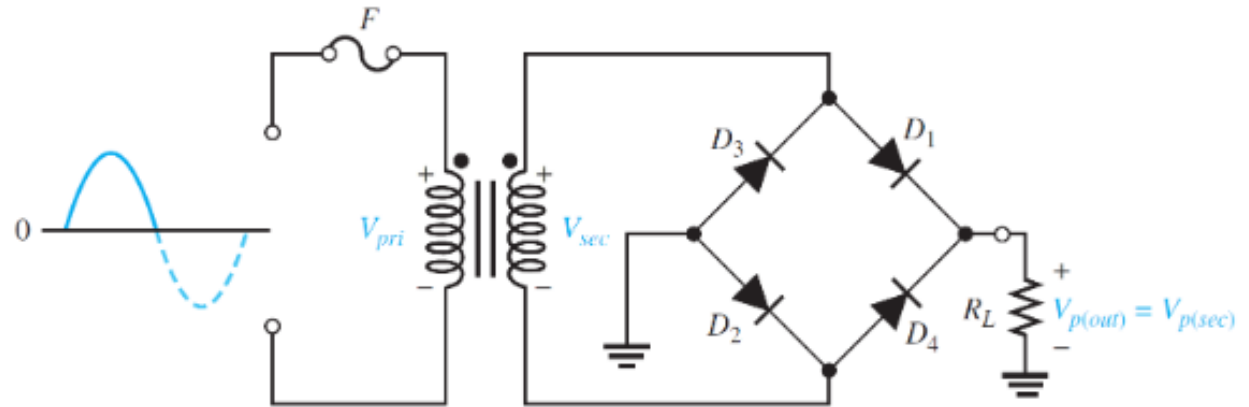
$$V_{p(out)} = V_{p(sec)} - 1.4$$



Onda completa Ponte

Ideal:

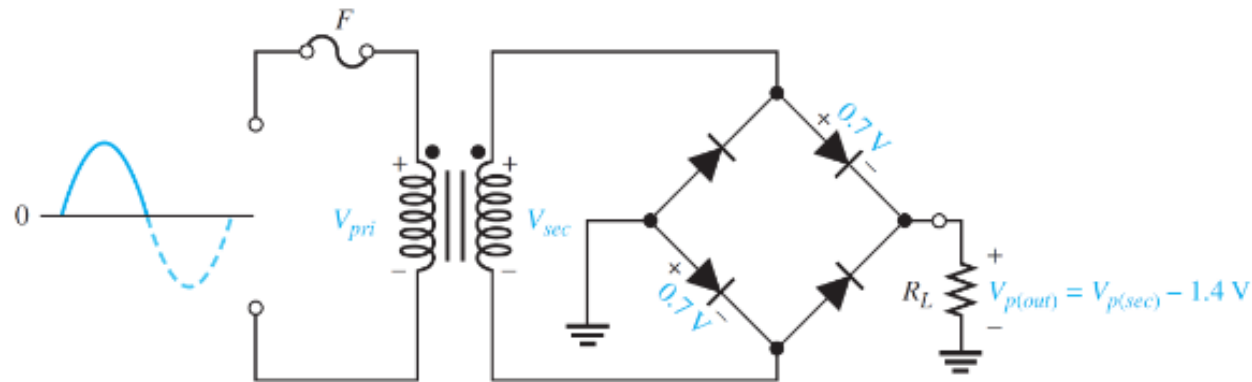
$$V_{p(out)} = V_{p(sec)}$$



(a) Ideal diodes

Practical:

$$V_{p(out)} = V_{p(sec)} - 1.4$$



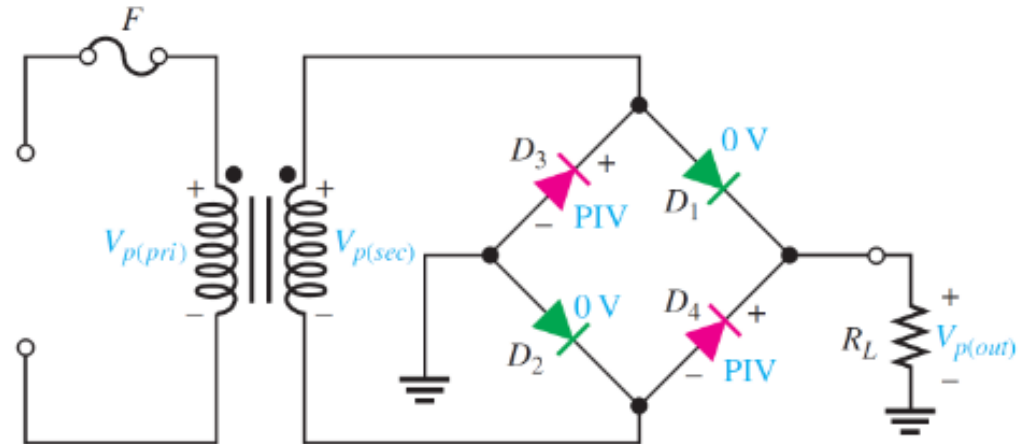
(b) Practical diodes (Diode drops included)

Onda completa Ponte

Peak Inverse Voltage

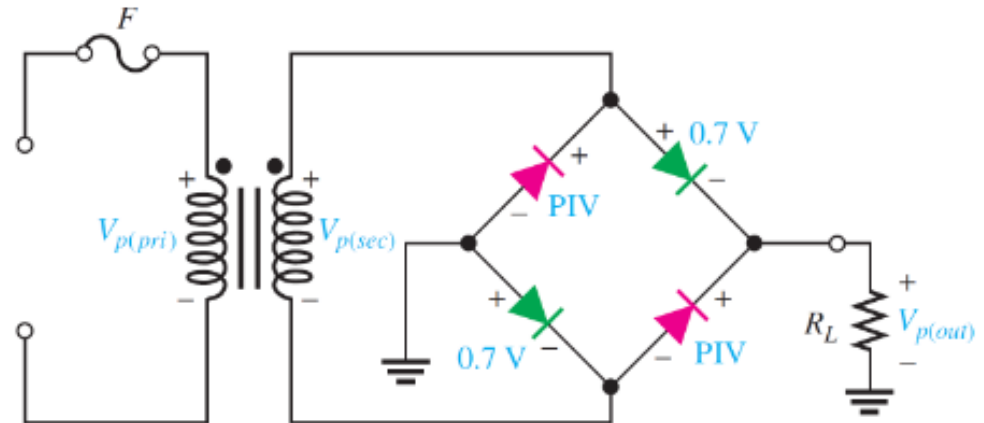
Ideal:

$$PIV = V_{p(out)}$$



Practical:

$$PIV = V_{p(out)} + 0.7$$



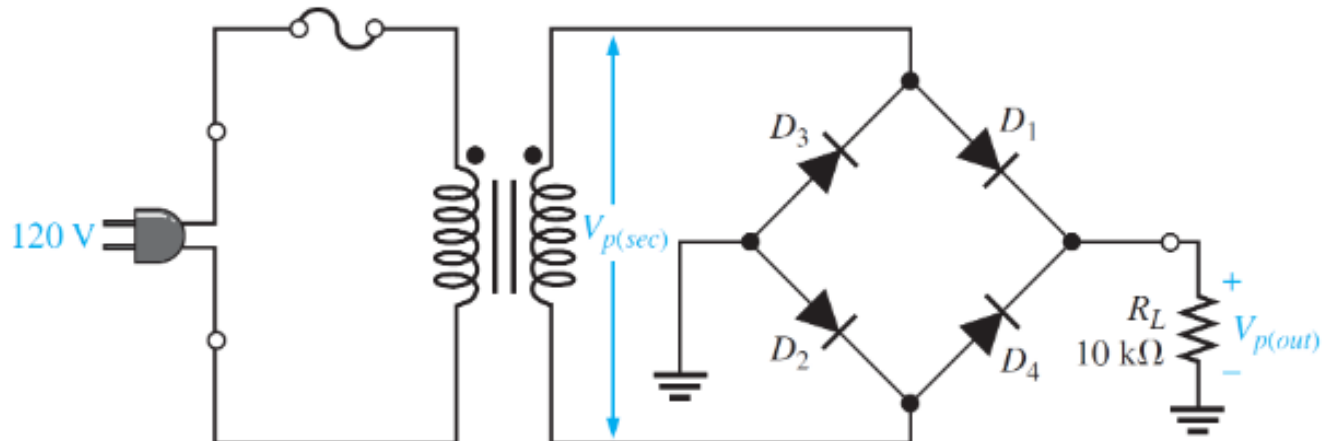
Onda completa Ponte

Example 5

Determine the peak output voltage for the bridge rectifier in Figure. Assuming the practical model, what PIV rating is required for the diodes? The transformer is specified to have a 12 V rms secondary voltage for the standard 120 V across the primary.

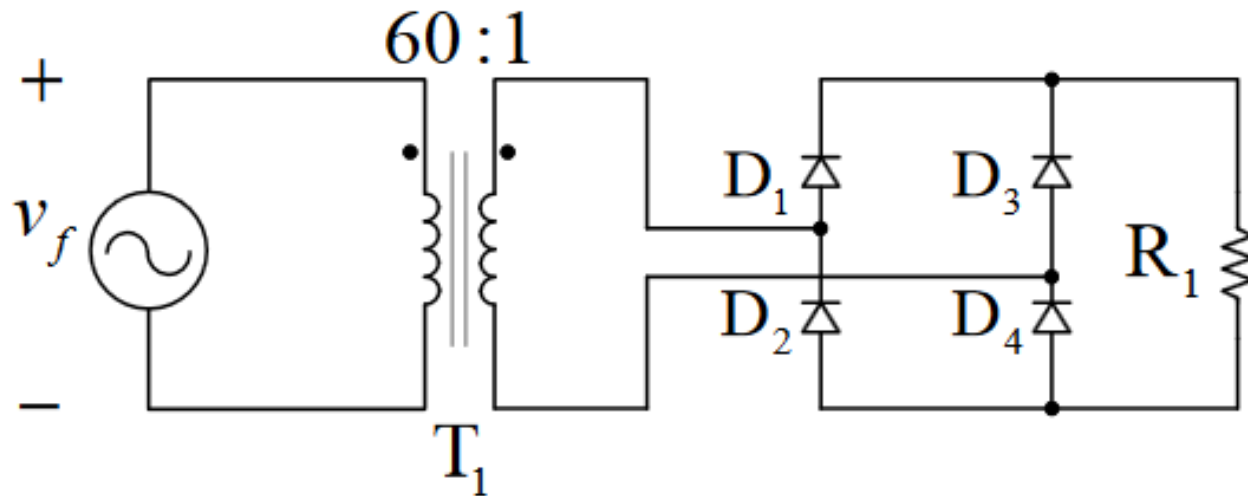
Solution

- The peak secondary voltage is : $V_{p(sec)} = \sqrt{2}V_{rms(sec)} = \sqrt{2} \times 12 = 17\text{ V}$
- The peak output voltage is: $V_{p(out)} = V_{p(sec)} - 1.4 = 15.6\text{ V}$
- The PIV rating for each diode is : $PIV = V_{p(out)} + 0.7 = 16.3\text{ V}$



Onda completa Ponte - Exercício

Considerando o circuito abaixo:



Considerando os dados ao lado, determine:

- Tensão eficaz no primário de T_1 ;
- Tensão eficaz no secundário de T_1 ;
- Tensão média na saída;
- Tensão de pico na saída;
- Tensão reversa sobre os diodos;
- Corrente média na saída.

$$v_f(t) = 311 \cdot \sin(377 \cdot t) \text{ V};$$

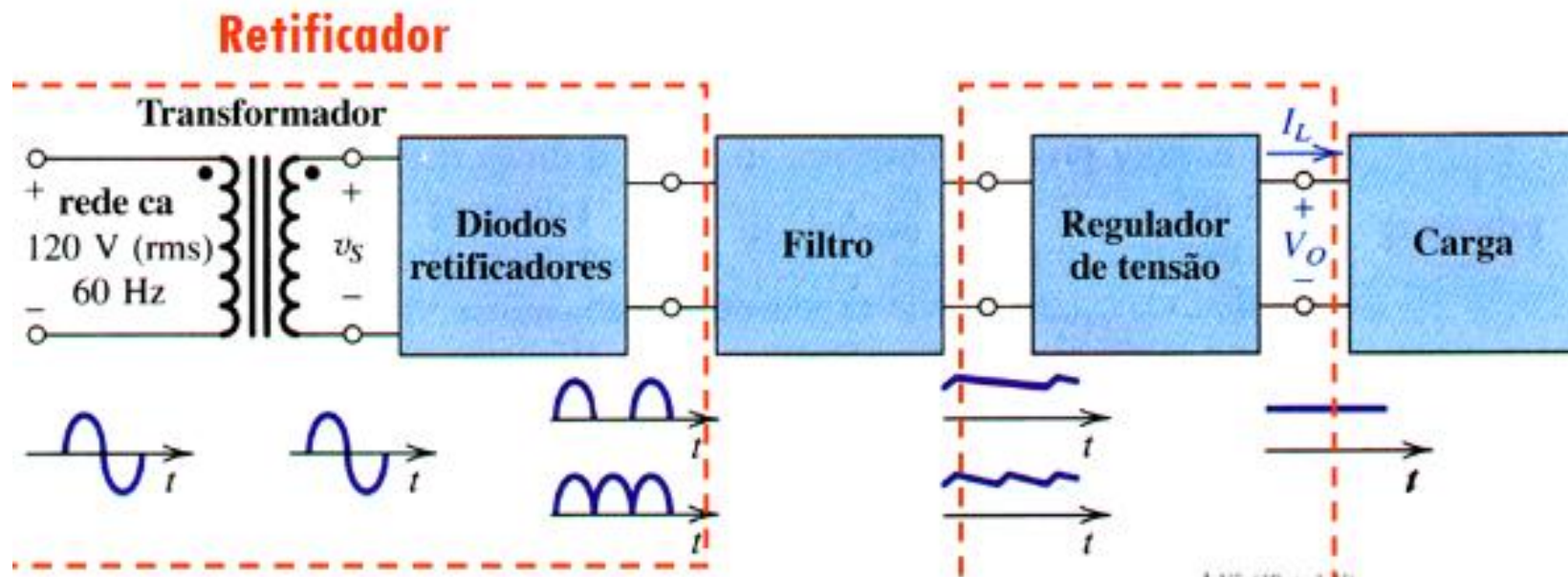
$$R_1 = 5 \, \Omega;$$

$$D_{1-4} = \text{ideais};$$

$$T_1 = 60:1.$$

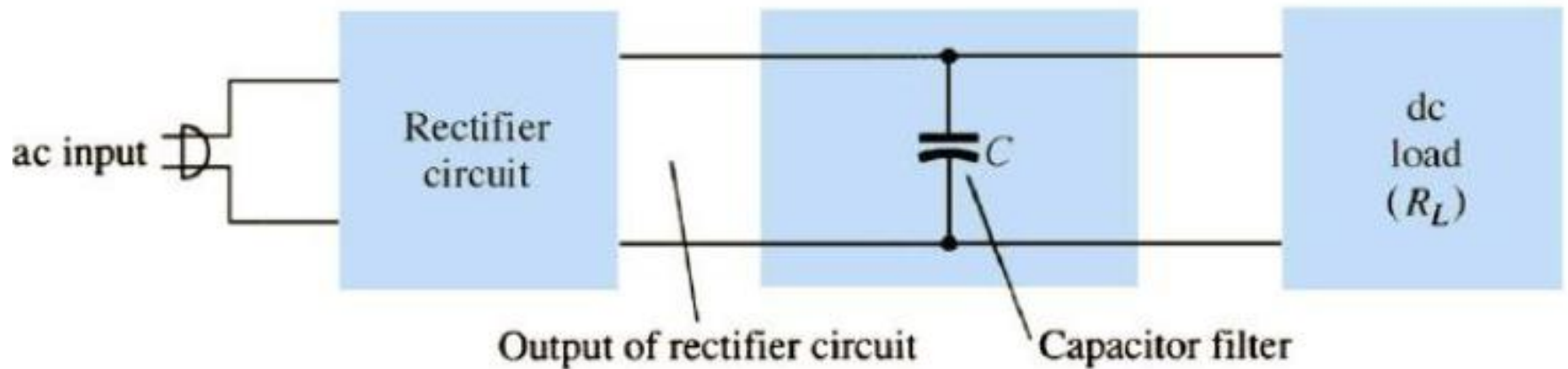
Retificadores

- Circuitos utilizados para conversão de CA/CC

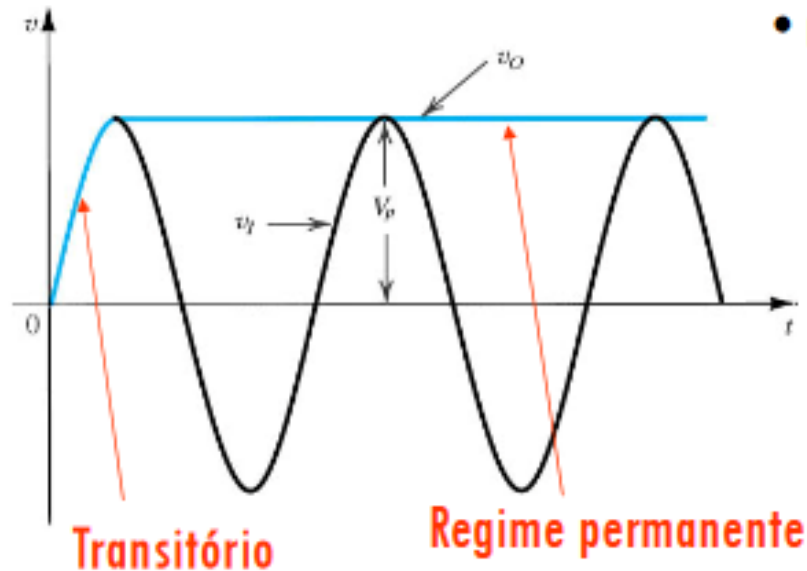
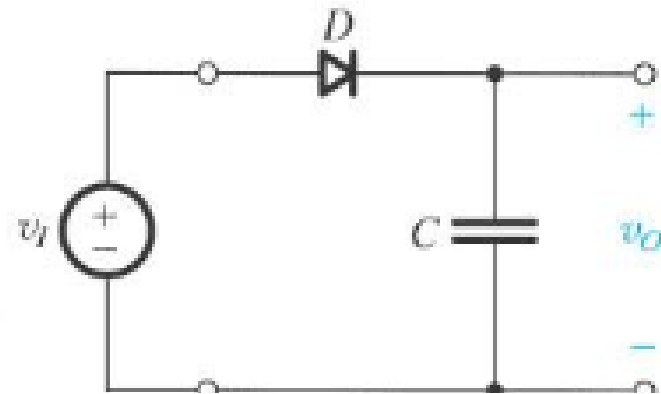


Filtro Capacitivo

Filtro com um único capacitor:

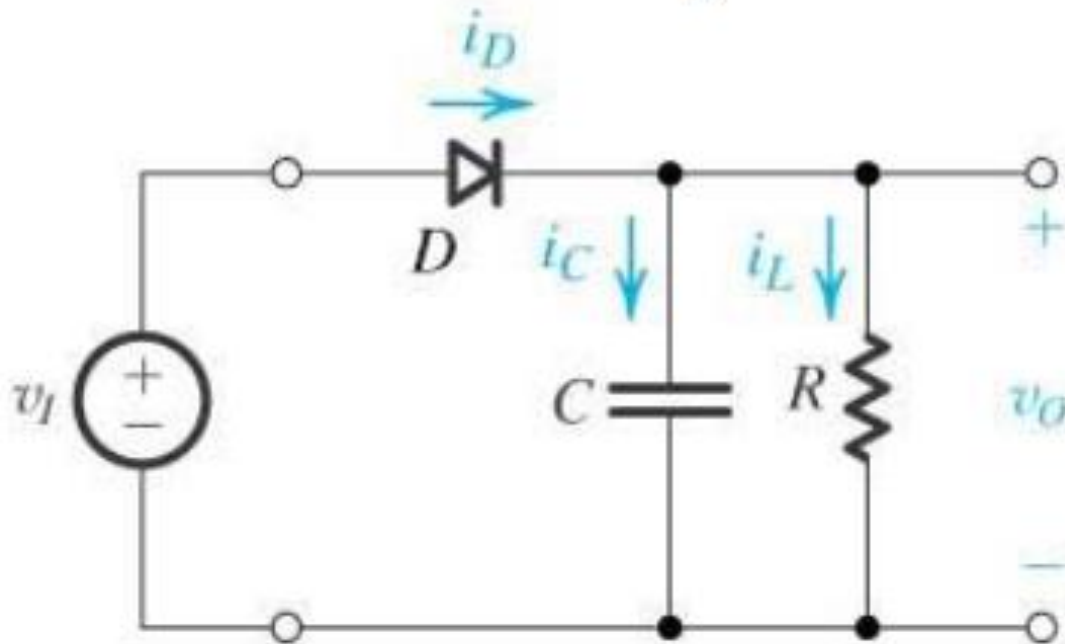


Filtro Capacitivo



Filtro Capacitivo

E com carga?



Atenção:

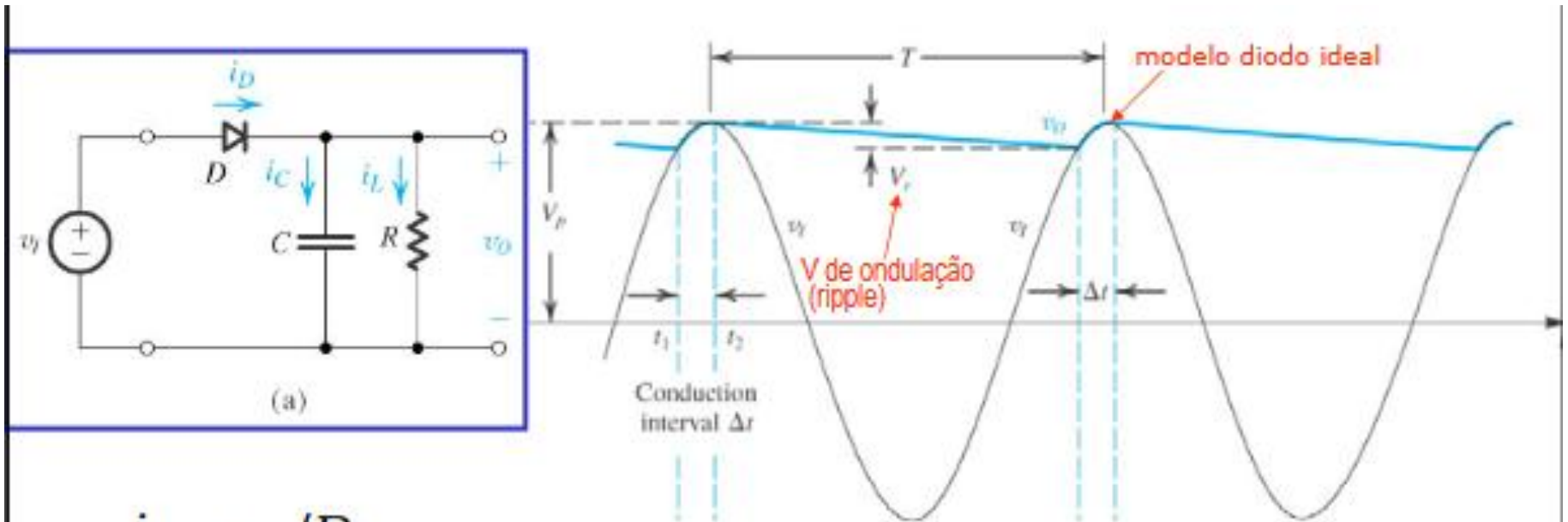
$$i_D = i_C + i_L$$

$$v_O = R_L i_L$$

$$v_O = \frac{1}{C} \int_0^T i_C dt$$

Quando o diodo conduz: $v_O(t) = v_I(t)$

Filtro Capacitivo



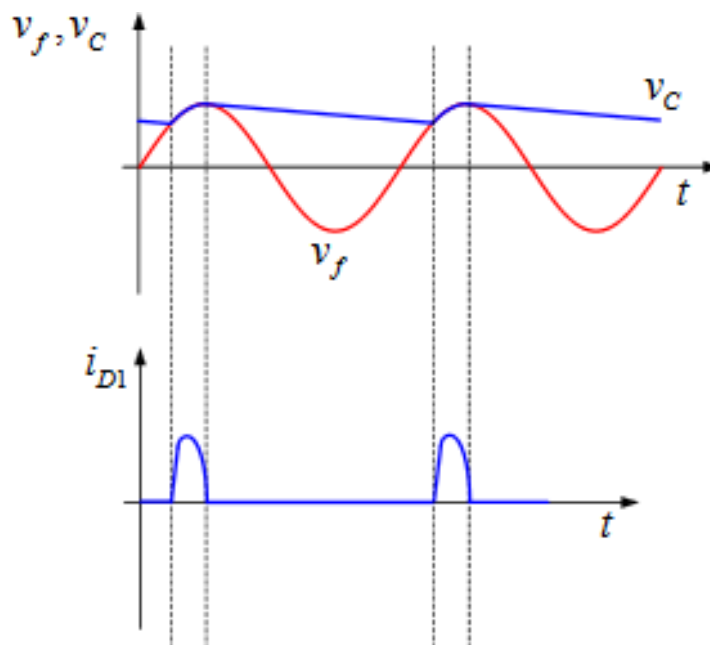
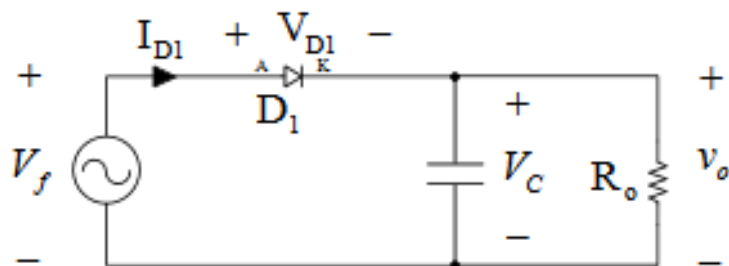
$$i_L = v_O / R_L$$

Quando D está conduzindo:

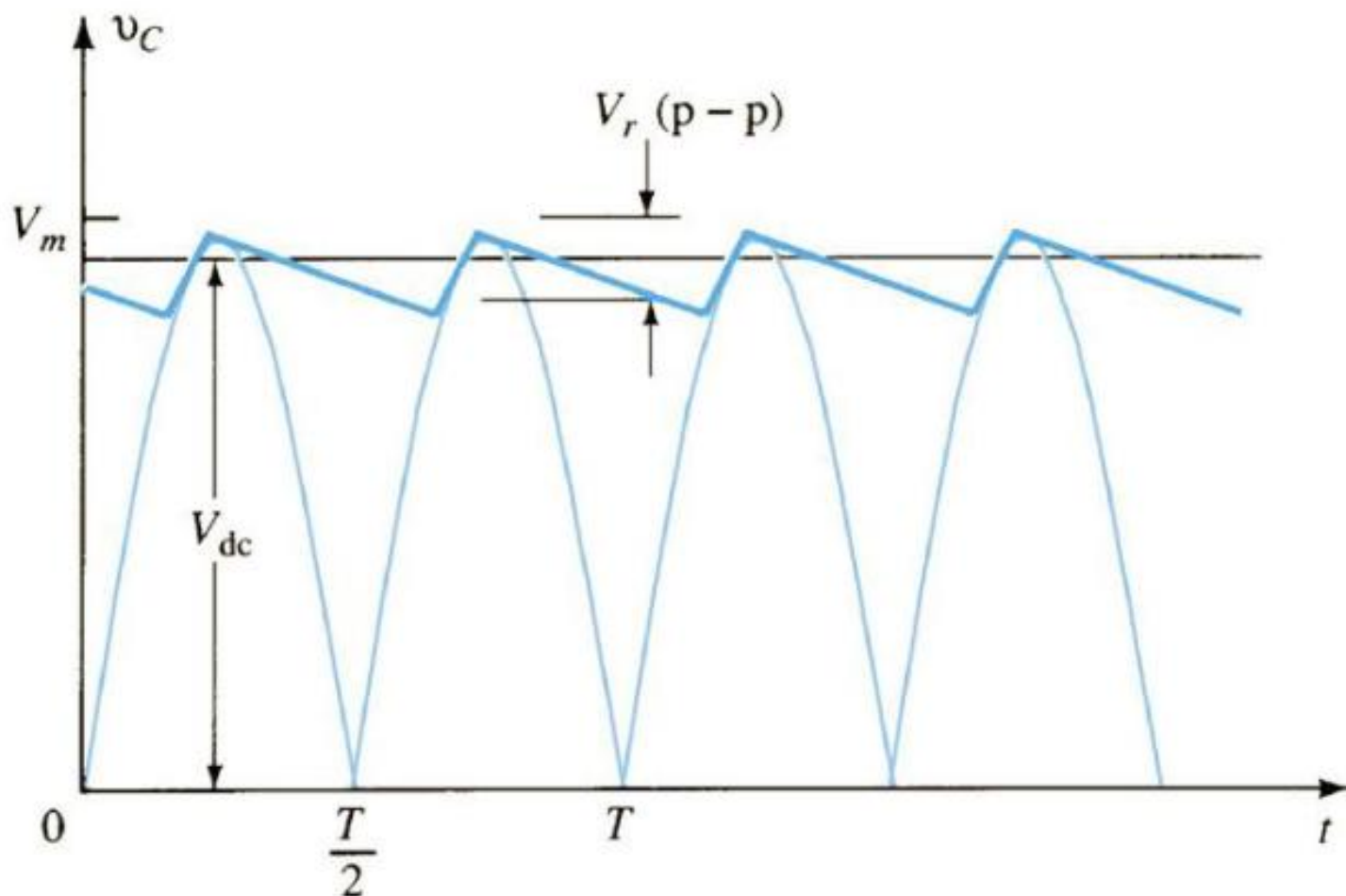
$$i_D = i_C + i_L$$

$$i_D = C \frac{dv_I}{dt} + i_L$$

Filtro Capacitivo – Tensão de ripple



Filtro Capacitivo – Tensão de ripple



Filtro Capacitivo

Determinando V_o (valor médio) e V_r

$$V_{O(médio)} = V_p - \frac{1}{2} V_r$$

Quando $CR \gg T$ (V_r pequeno):

$$i_{L(médio)} = I_L = \frac{V_p}{R}$$

Para $CR \gg T$:

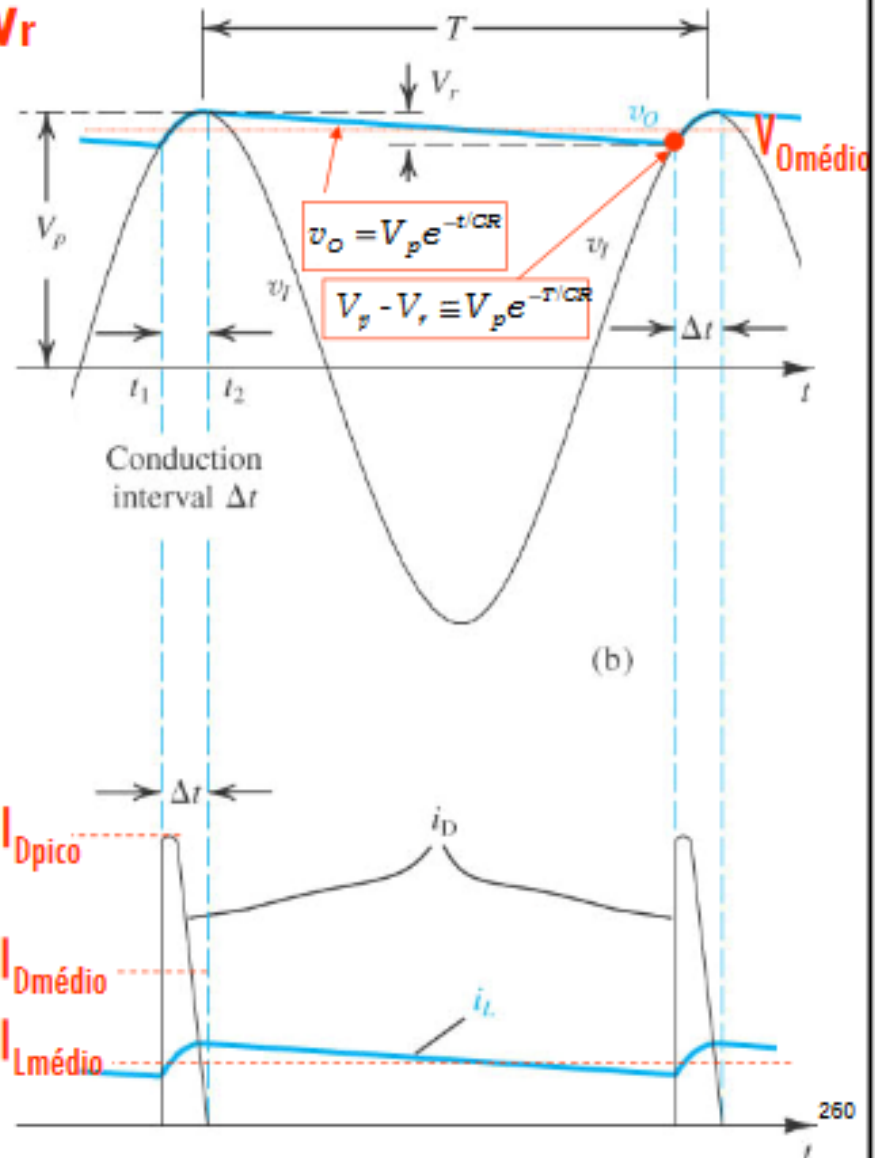
$$e^{-T/CR} \cong 1 - T/CR$$

$$V_p - V_r \cong V_p(1 - T/CR)$$

e portanto:

$$V_r \cong V_p \frac{T}{CR} = \frac{V_p}{fRC}$$

ou $V_r = \frac{I_L}{fC}$



Filtro Capacitivo - Expressões

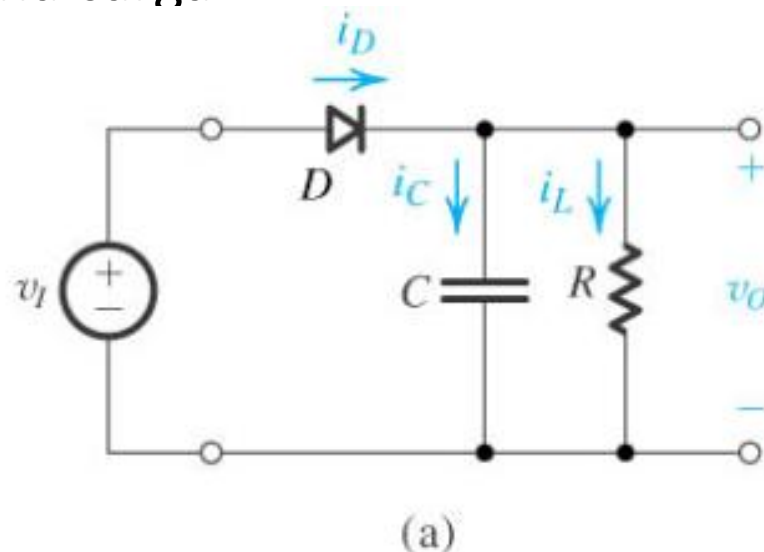
- Resumo - Principais fórmulas

$$V_{DC} = V_p - \frac{V_r}{2}$$

$$V_r = \frac{V_p}{fRC}$$

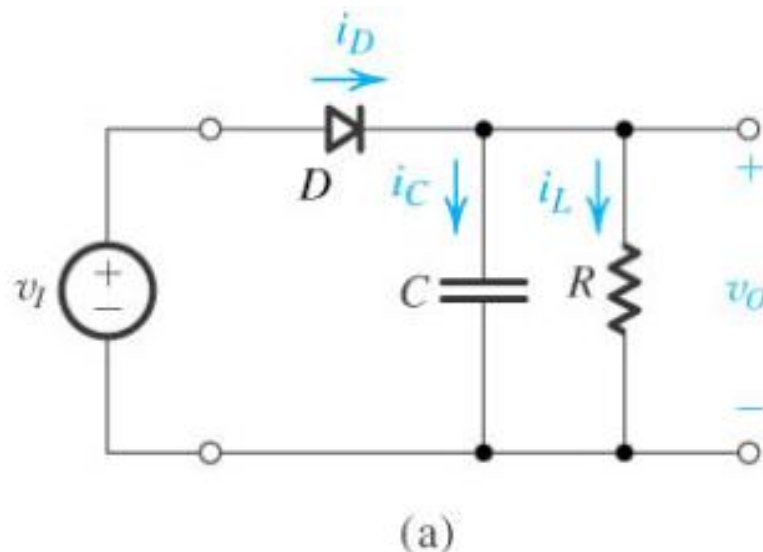
Filtro Capacitivo - Exercícios

- Considere um retificador meia onda com v_i de 100V pico, frequência de 60Hz, C 100 μ F e R 10k. Calcule:
- A) A tensão de pico na saída
- B) A tensão de ripple .
- C) A tensão média na carga



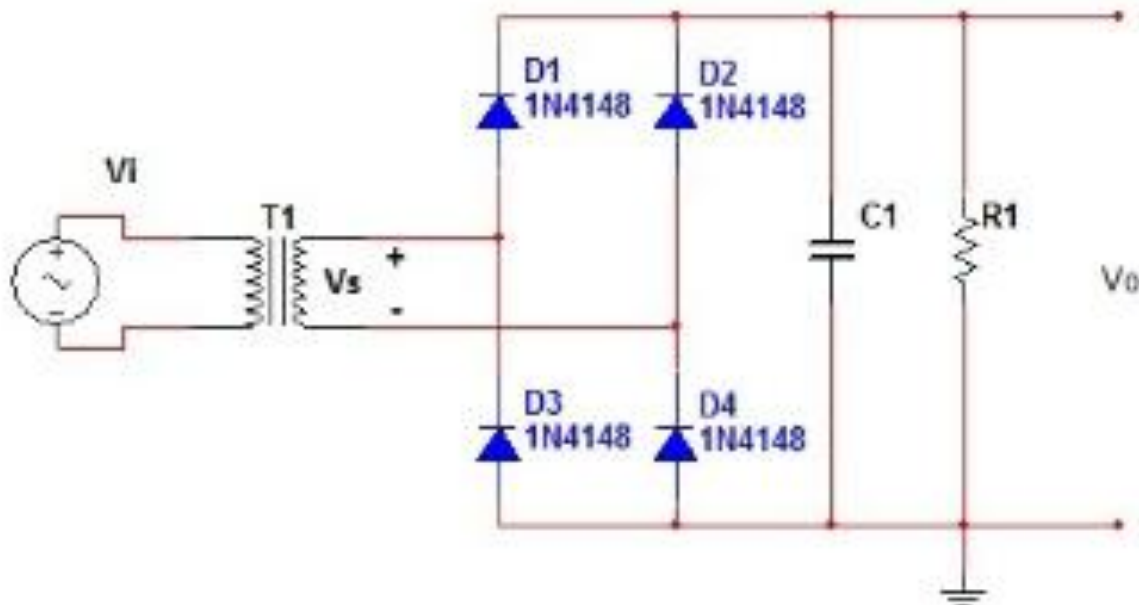
Filtro Capacitivo - Exercícios

- **Simule** o circuito anterior e obtenha:
- A) As tensões calculadas
- B) A forma de onda da tensão de saída.
- C) A forma de onda da tensão de ripple
- D) A forma de onda da corrente no diodo



Filtro Capacitivo - Exercícios

- **Repita** o exercício anterior para um retificador de onda completa em ponte. Lembre-se que v_i é 100V pico, frequência de v_i 60Hz, $C(C1)$ 100 μ F e $R(R1)$ 10k.



Disciplina: Eletrônica Básica

Aula_5 - Diodos

Ambiente Blackboard